⚡ LAST-MINUTE REVISION

High-Yield Summary: Definitions, Comparison Tables & Attack Lifecycles

🎯 **PURPOSE:** Revise the complete Flood Attack topic in under 10 minutes. Memorize the 10 must-remember points, distinguish common exam distractors, and master the 5-stage lifecycle.

1. ONE-LINE DEFINITIONS

Concept	Exam Definition (One-Liner)
Flood Attack	An offensive tactic sending massive request volumes to overwhelm network capacity or CPU interrupts.
DoS vs DDoS	DoS uses a single attacking node; DDoS uses a coordinated distributed network (botnet) of nodes.
ICMP Protocol	Network Layer (OSI Layer 3) diagnostic protocol used by utility tools like ping and traceroute.
Ping Utility	Sends an ICMP Echo Request (Type 8) expecting an ICMP Echo Reply (Type 0) to measure reachability and RTT.
PPS (Packets/Sec)	Packet-rate throughput metric; flooding overwhelms CPU interrupt handlers even with tiny packet payload sizes.
Availability	The 'A' in the CIA Triad; ensuring systems, applications, and network resources remain accessible to authorized users.
Rate Limiting	Defensive policy restricting the maximum number of requests a host or network can process in a set time window.
Firewall Filtering	Inspection rule set evaluating incoming packets by IP, protocol, or port to permit or drop unauthorized packets.

2. ESSENTIAL DIFFERENCES & COMPARISON TABLE

Attack Type	Layer	Target Resource	Mechanism	Primary Defense
ICMP Flood	Layer 3	CPU Interrupts / Bandwidth	Continuous Echo Request bursts	iptables rate-limiting / Drop ICMP
SYN Flood	Layer 4	TCP Connection Table (RAM)	Incomplete 3-way handshakes	SYN Cookies / TCP connection limits
UDP Flood	Layer 4	Socket processing / ICMP errors	Random port bursts forcing ICMP unreachable	Upstream traffic filtering / UDP drop
HTTP Flood	Layer 7	Web Server CPU & Database	Resource-heavy GET/POST requests	Web App Firewall (WAF), CAPTCHA, CDN caching

3. THE FIVE-STAGE FLOOD ATTACK LIFECYCLE

1. **Stage 1: Normal Baseline** — System operates with healthy PPS (<100 PPS), low CPU (<10%), and sub-millisecond response latency.
2. **Stage 2: Traffic Influx** — Attacker transmits rapid bursts; incoming PPS spikes abruptly to 25,000–50,000+ packets/sec.
3. **Stage 3: Resource Depletion** — Target kernel experiences CPU SoftIRQ lockup or memory exhaustion; genuine requests drop.
4. **Stage 4: Mitigation Engagement** — Firewall/WAF rules or upstream scrubbers drop offending traffic; rate limiters take effect.

5. **Stage 5: Service Recovery** — Traffic throughput normalizes, kernel queues clear, and legitimate user availability is restored.

4. IMPORTANT METRICS & DIAGNOSTIC INTERPRETATION

Metric	Normal Baseline	Flood Attack State	Diagnostic Meaning
PPS	10 – 100 PPS	20,000 – 60,000+ PPS	High PPS indicates packet volumetric surge, exhausting kernel SoftIRQs.
CPU %	2% – 12%	90% – 100%	Processor saturated by operating system network stack context switching.
Latency (RTT)	< 5 ms	> 2,000 ms / Timeout	Requests queuing up; buffer bloat and server unable to reply in time.
Packet Loss	0%	60% – 100%	Network queues full; router/server NIC silently discards legitimate traffic.

5. TEN MUST-REMEMBER EXAM POINTS

- 1. **Availability is the target:** Floods disrupt accessibility, NOT confidentiality or data integrity.
- 2. **Encryption does NOT stop floods:** HTTPS/TLS secures message privacy, but encrypted packets still consume bandwidth and CPU.
- 3. **ICMP Echo Request is Type 8; Echo Reply is Type 0.**
- 4. **Rate limiting drops excess packets:** Genuine traffic continues to be served up to the configured threshold.
- 5. **Bandwidth saturation vs Resource exhaustion:** Pipe full = Bandwidth issue; Pipe open but CPU at 100% = Resource exhaustion.
- 6. **False Positives:** Legitimate flash sales or exam results days can mimic floods; defenses must avoid banning valid users.
- 7. **Layered Defense (Defense in Depth):** No single tool solves DDoS; combine edge scrubbing, firewalls, load balancers, and WAFs.
- 8. **iptables limit module syntax:** `-m limit --limit 1/s --limit-burst 5 -j ACCEPT`.
- 9. **SYN Flood exploits TCP 3-way handshake:** Resolved using SYN cookies without keeping state for embryonic connections.
- 10. **Docker Lab Safety:** Always execute stress tests inside an isolated bridge network, NEVER across public internet addresses.

6. FREQUENTLY CONFUSED CONCEPTS

❌ **Confusing DoS with Data Breach:** A server crashing from a ping flood does NOT mean attacker stole passwords or database tables.

❌ **Confusing Rate Limiting with Full Blocking:** Full blocking rejects 100% of traffic; rate limiting permits normal traffic while trimming the peak.

❌ **Confusing High Latency with Guaranteed Attack:** Poor Wi-Fi or overloaded ISP routing causes latency too; verify packet rates and CPU telemetry before declaring an incident.

7. QUICK SELF-TEST (EXAM PRACTICE)

Q1: Why does ping flood crash a server if each packet is only 64 bytes?
Ans: The sheer volume (tens of thousands PPS) causes kernel SoftIRQ interrupt starvation, consuming 100% CPU on packet headers.

Q2: If an e-commerce website experiences a traffic spike during Black Friday, how do you distinguish it from an HTTP flood?
Ans: Inspect HTTP request patterns, user-agent variety, referrer headers, session cookies, and geographic diversity vs repeated scripted URLs.

Q3: Which defense is best against network pipe saturation?
Ans: Upstream Anycast DDoS scrubbing (Cloudflare, AWS Shield) before packets reach the local data center pipe.